Physics 1100: Final Examination

Your Name:
(Dated: May 3, 2005, 8-10 am)

Remember: you must provide an **appropriate diagram** to get a full credit for the problem!
Gravitational acceleration is $g = 9.80$ m/s^2.

P1. A record of travel along a straight path is as follows:
(1) Start from rest with constant acceleration $a_1 = 2.77$ m/s^2 for $t_1 = 15.0$ s.
(2) Constant velocity v_2 for the next $t_2 = 2.05$ min.
(3) Constant negative acceleration $a_3 = -9.47$ m/s^2 for $t_3 = 4.39$ s.
5 pts (a) What is velocity v_2 for the second part (of duration t_2) for this trip?
10 pts (b) What is the total displacement Δx for the complete trip?
5 pts (c) What is the average speed $\bar{v}$ for the complete trip?

P2. Two people pull on a stubborn mule, as seen from a helicopter in Figure 1, $F_1 = 120$ N at $\theta_1 = 60°$ from positive x-direction, and $F_2 = 80$ N at $\theta_2 = 75°$ from *negative* x-direction.
10 pts (a) Find the single force $\vec{F}_R = \vec{F}_1 + \vec{F}_2$ that is equivalent to the two forces shown (both its magnitude and direction).
5 pts (b) Find the force $\vec{F}_3$ that a third person would have to exert on the mule to make the net force $\sum \vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$ equal to zero, $\sum \vec{F} = 0$.

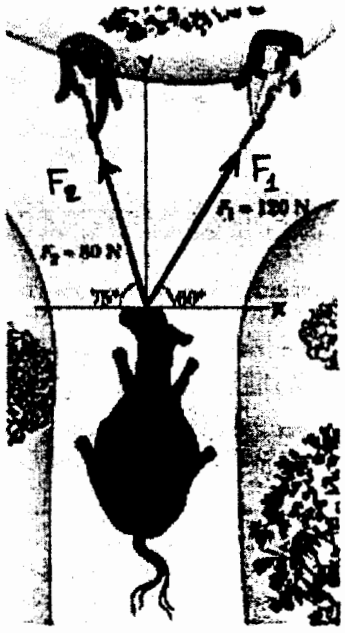


FIG. 1: Problem P2.

P3. A student decides to move a box of books into her dormitory room by pulling on a rope attached to the box. She pulls with a force $F = 80$ N at an angle $\theta = 25.0°$ above the horizontal. The box has mass $m = 25.0$ kg, and the coefficient of kinetic friction between box and floor is $\mu_k = 0.300$.
5 pts (a) Draw a free-body diagram for the box.
5 pts (b) Find the acceleration a of the box.
10 pts (c) Calculate work done by each of the forces present in the problem during a $\Delta x = 4.00$ m displacement of the box, and the total work done on the box.
5 pts (d) Assuming that the box started from rest, find its speed v_f at the end of $\Delta x = 4.00$ m displacement.

P4. A $m_1 = 65.0$ kg person throws a $m_b = 1.20$ kg basketball forward with a speed $v_b = 20.0$ m/s. A second person, initially at rest and of mass $m_2 = 55.0$ kg, catches the basketball. Both people are on skates so that the friction is negligible.

5 pts (a) What is the velocity v_1 of the first person, initially at rest, after throwing the ball?

5 pts (b) What is the velocity v_2 of the second person after he catches the ball?

5 pts (c) What best describes *relative* motion of these two people after the basketball is exchanged? (Circle one answer and provide your arguments in its favor in your exam bluebook.)

(1) They move towards each other along the line connecting them.

(2) They move away from each other along the line connecting them.

(3) Their relative motion is too complicated to describe in few words.

P5. A block of mass $m = 0.3$ kg slides on a semicircular frictionless track of radius $R = 1.2$ m (Figure 2). If it starts from rest at position A,

5pts (a) What is its speed v_B at point B?

5 pts (b) What is its speed v_C at point C?

Assume now that an identical semicircular track is made rough enough so that a uniform solid disk of mass $M_{\text{disk}} = 0.3$ kg and radius $r_{\text{disk}} = 0.1$ m can roll down the track without slipping. Given that the disk starts from rest at point A,

5 pts (c) Find disk's speed at point C.

5 pts (d) What is disk's rotational kinetic energy K_r at that point?

Hint: moment of inertia of the disk is $I_{\text{disk}} = \frac{1}{2} M_{\text{disk}} r_{\text{disk}}^2$.

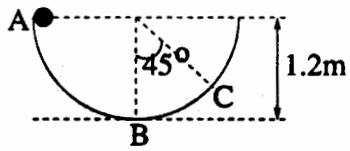


FIG. 2: Problem P5.

P6. A uniform iron bar of length $L = 4.0$ m is held perpendicular to the wall by a wire joined from the end of the bar to the wall as shown in Figure 3.

10 pts (a) What is the tension T in the wire if the iron bar weights $w = 400$ N and the wire is $d = 5.0$ m long?

10 pts (b) What is the reaction force $\vec{P}$ from the wall on the bar? (you need to find its magnitude P and direction, or, equivalently, its x and y components.)

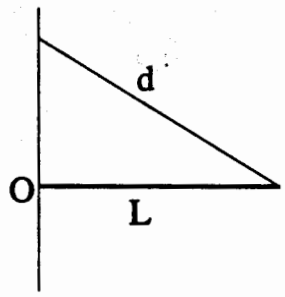
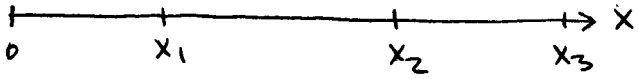


FIG. 3: Problem P6.

Bonus P7. If a force of 4.0 N is needed to open a 1.0-m wide door when applied at the edge opposite the hinges, what force must be applied to open the door if you push against the door 10 cm from the hinged side? **(10 pts.)**

Final exam Physics 1100, May 3, 2005. Starych

P1



$a_1 = 2.77\,m/s^2$ $a_2 = 0$ $a_3 = -9.47\,m/s^2$

$t_1 = 15.0\,s$ $t_2 = 2.05\,min$ $t_3 = 4.39\,s$

a) v_2 = final velocity at $t_1 = 15.0\,s$

$$\Rightarrow v_2 = \underset{0}{\underset{\shortparallel}{v_i}} + a_1 t_1 = 2.77\,m/s^2 \cdot 15.0\,s = \boxed{41.55\,m/s}$$

b) $\Delta x = (x_1 - 0) + (x_2 - x_1) + (x_3 - x_2)$

$$x_1 = \underset{0}{\underset{\shortparallel}{v_i}} t_1 + \frac{1}{2} a_1 t_1^2 = \frac{1}{2}\, 2.77\,m/s^2\,(15.0\,s)^2 = 311.6\,m$$

$$x_2 - x_1 = v_2 t_2 = 41.55\,m/s \cdot 123\,s = 5110.7\,m$$

$$t_2 = 2.05\,min = 2.05\,min \cdot \frac{60\,s}{min} = 123\,s$$

$$x_3 - x_2 = v_2 t_3 + \frac{1}{2} a_3 t_3^2 =$$

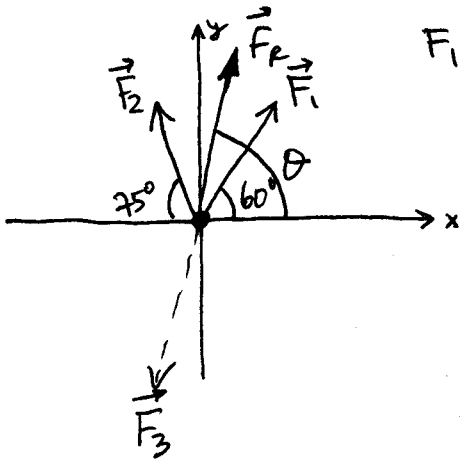
$$= 41.55 \cdot 4.39 - \frac{1}{2}\, 9.47 \cdot (4.39)^2 = 182.4 - 91.25 = 91.1\,m$$

$$\Rightarrow \Delta x = 311.6 + 5110.7 + 91.1 = \boxed{5513\,m}$$

c) average speed $\bar{v} = \frac{\Delta x}{\Delta t} = \frac{5513\,m}{142.39\,s} = \boxed{38.7\,m/s}$

$$\Delta t = 15.0 + 123 + 4.39 = 142.39\,s$$

P2



$F_1 = 120\,N$, $F_2 = 80\,N$

(a) $\vec{F}_R = \vec{F}_1 + \vec{F}_2$

$$\vec{F}_1 = (120 \cdot \cos 60°,\ 120 \sin 60°) = (60,\ 103.9)\,N$$

$$\vec{F}_2 = (-80 \cos 75°,\ 80 \sin 75°) = (-20.7,\ 77.3)\,N$$

$\Rightarrow \vec{F}_R = (60 - 20.7,\ 103.9 + 77.3) = (39.3,\ 181.2)\ N$

Magnitude $F_R = |\vec{F}_R| = \sqrt{(39.3)^2 + (181.2)^2} = 185.4 = \boxed{185\,N}$

Direction $\theta = \tan^{-1}\dfrac{F_{Ry}}{F_{Rx}} = \tan^{-1}\dfrac{181.2}{39.3} = \boxed{77.8^\circ}$ from positive x-axis

(b) $\vec{F}_3$ such that $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 0$

$\Rightarrow \vec{F}_3 = -\vec{F}_1 - \vec{F}_2 = -\vec{F}_R$

$\Rightarrow \vec{F}_3 = (-39.3,\ -181.2)\ N$

Magnitude $F_3 = F_R = \boxed{185\,N}$

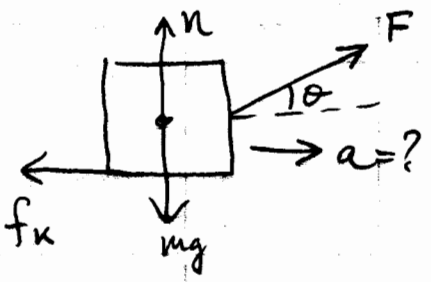
Direction: opposite to $\vec{F}_R \Rightarrow \theta_3 = 180^\circ + 77.8^\circ = \boxed{257.8^\circ}$ from $x>0$

or -102.2° from $x>0$

or -77.8° from $\underline{x<0}$

$\boxed{P3}$

(a)



$F = 800N,\ \theta = 25.0^\circ$

$m = 25.0\,kg,\ \mu_k = 0.300$

$\Delta x = 4.00\,m$

(b) Newton's 2nd law: $\begin{cases} F\cos\theta - f_k = ma \\ F\sin\theta + n - mg = 0 \end{cases}$

$\Rightarrow n = mg - F\sin\theta = 25 \cdot 9.8 - 80\sin 25^\circ = 211.2\ N$

$\Rightarrow f_k = \mu_k n = 0.300 \cdot 211.2 = 63.4\,N$

$\Rightarrow ma = 80\cos 25^\circ - 63.4 = 9.1\,N \Rightarrow a = \dfrac{9.1N}{25kg} = \boxed{0.364\,m/s^2}$

(c) $W_n = W_g = 0$ because they are orthogonal to $\vec{\Delta x}$.

$$W_F = F\,\Delta x \cos\theta = 80\,N \cdot \cos 25^\circ \cdot 4.00\,m = \underline{290\ J}$$

$$W_f = f_k\,\Delta x \cos 180^\circ = -f_k\,\Delta x = -63.4\,N \cdot 4.00\,m = \underline{-254\ J}$$

$$W_{total} = 290 - 254 = \boxed{36\ J}$$

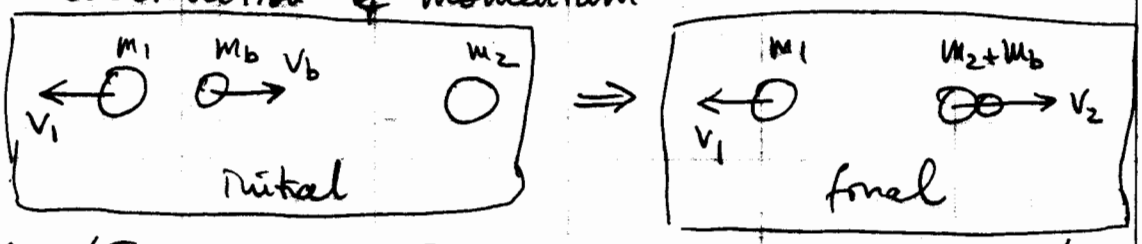
(d) Work – KE Theorem: $W_{total} = KE_f - KE_i$

$$KE_i = \tfrac{1}{2} m v_i^2 = 0 \text{ because } v_i = 0$$

$$\Rightarrow \tfrac{1}{2} m v_f^2 = W_{total} = 36\ J$$

$$v_f = \sqrt{\frac{2\,W_{total}}{m}} = \sqrt{\frac{2 \cdot 36}{25}} = \sqrt{2.88} = \boxed{1.70\ m/s}$$

P4 Conservation of momentum



$m_1 = 65.0\ kg$, $m_2 = 55.0\ kg$, $m_b = 1.20\ kg$, $v_b = 20.0\ m/s$

(a) $m_1 v_{1i} + m_b v_{bi} = 0 = m_1 v_1 + m_b v_b$

$$\Rightarrow v_1 = -\frac{m_b v_b}{m_1} = -\frac{1.20 \cdot 20}{65.0} = \boxed{-0.369\ m/s}$$

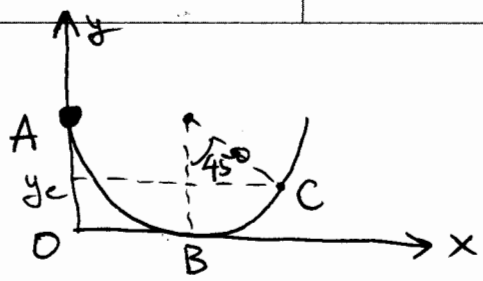
(b)

$$m_2 \underset{\substack{\parallel \\ 0}}{v_{2i}} + m_b v_b = (m_2 + m_b)\, v_2$$

$$\Rightarrow v_2 = \frac{m_b v_b}{m_2 + m_b} = \frac{1.2 \cdot 20}{55 + 1.2} = \boxed{0.427\ m/s}$$

(c) (2) They move away from each other: $v_1 < 0$, $v_2 > 0$.

P5



choose $y_B = 0$

$\Rightarrow y_A = R = 1.2\ m$

$m = 0.3\ kg$

$V_A = 0$

(a) $E_i = mgy_A = mgR$

$E_f = mg\underset{0}{y_B} + \frac{1}{2}mv_B^2 = \frac{1}{2}mv_B^2$

$\Rightarrow \frac{1}{2}mv_B^2 = mgR \Rightarrow v_B = \sqrt{2gR} = \boxed{4.85\ m/s}$

(b) $E_c = mgy_c + \frac{1}{2}mv_c^2$

$y_c = R - R\sin 45° = 1.2(1-\cos 45°) = 0.35\ m$

$\Rightarrow \frac{1}{2}mv_c^2 + mgy_c = mgy_A$

$\frac{1}{2}v_c^2 = g(y_A - y_c) = g(R - R(1-\cos 45)) = gR\sin 45°$

$v_c = \sqrt{2gR\cos 45°} = \boxed{4.08\ m/s}$

(c) Disk rolling down from A to C.

$I_{disk} = \frac{1}{2}Mr^2$

$E_c = \frac{1}{2}Mv_c^2 + \frac{1}{2}I\omega_c^2 + Mgy_c$

$\omega_c = \frac{v_c}{r} \Rightarrow E_c = \frac{1}{2}Mv_c^2 + \frac{1}{2}\left(\frac{1}{2}M\cancel{r^2}\right)\frac{v_c^2}{\cancel{r^2}} + Mgy_c$

$E_c = \frac{3}{4}Mv_c^2 + Mgy_c$

$E_A = Mgy_A$

$\frac{3}{4}Mv_c^2 + Mgy_c = Mgy_A$

$\frac{3}{4}v_c^2 = g(y_A - y_c) = gR\sin 45°$

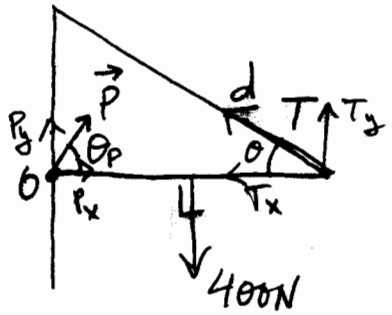
$v_c = \sqrt{\frac{4}{3}gR\cos 45°} = \boxed{3.33\ m/s}$

(d) $V_c = 3.33\ m/s \Rightarrow \omega_c = \frac{V_c}{r} = \frac{3.33\ m/s}{0.1\ m} = 33.3\ \frac{rad}{s}$

$$K_r = \frac{1}{2} I \omega_c^2 = \frac{1}{2}\left(\frac{1}{2} M r^2\right)\left(\frac{V_c}{r}\right)^2 = \frac{1}{4} M V_c^2$$

$$= \frac{1}{4}\, 0.3 (3.33)^2 = \boxed{0.83\ J}$$

P6



(1) $P_x - T_x = 0 \Rightarrow P_x = T\cos\theta$

(2) $P_y + T_y - 400 = 0$

$\Rightarrow P_y = 400 - T\sin\theta$

(3) $\sum \tau_o = -400 \cdot \frac{L}{2} + L \cdot T_y = 0$

$\Rightarrow T_y = 200\ N \Rightarrow T = \frac{200\ N}{\sin\theta}$

$d\cos\theta = L \Rightarrow \cos\theta = \frac{L}{d} = \frac{4}{5} = 0.8$

$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{0.36} = 0.6$

(a) $\Rightarrow T = \frac{200\ N}{0.6} = \boxed{333\ N}$ $(\theta = 36.8°)$

(b) $P_y = 400 - T_y = \boxed{200\ N}$

$P_x = T\cos\theta = 333\ N \cdot 0.8 = \boxed{267\ N}$

Magnitude $P = \sqrt{P_x^2 + P_y^2} = \sqrt{200^2 + 267^2} = 333.6 = \boxed{334\ N}$

direction $\theta_p = \tan^{-1}\left(\frac{P_y}{P_x}\right) = \tan^{-1}\left(\frac{200}{267}\right) = \boxed{36.8°}$

BONUS

P7 $\tau = 4.0\ N \cdot 1.0\ m = 4.0\ N \cdot m$ torque required to open the door.

$\Rightarrow \tau = F \cdot 0.1\ m = 4\ N \cdot m$

$\Rightarrow \boxed{F = 40\ N}$ if applied at $x = 0.1\ m$ from the hinges.